ELEC70070
EE9-SO20

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE/EIE PART IV: MEng and ACGI

INFORMATION THEORY

Friday, 9 May 2025, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Information for students

Notation:

(a) Random variables are shown in Tahoma font. x, $\mathbf{x}$, $\mathbf{X}$ denote a random scalar, vector and matrix respectively.

(b) The size of a set A is denoted by $|A|$.

(c) By default, the logarithm is to the base 2.

(d) $\oplus$ denotes the exclusive-or operation, or modulo-2 addition.

(e) "i.i.d." means "independent identically distributed".

(f) $H(\cdot)$ is the entropy function.

(g) $C(x) = \frac{1}{2}\log_2(1+x)$ is the capacity function for the Gaussian channel in bits/channel use.

The Questions

1. Fundamentals of Information Theory (25 marks).

   a) A discrete random variable X takes values {0, 1, 2} with probabilities $P(X = 0) = 0.5$, $P(X = 1) = 0.3$, and $P(X = 2) = 0.2$. Compute the entropy $H(X)$.

   [6]

   b) Let X and Y be two binary random variables with joint distribution: $P(X=0, Y=0) = 0.1$, $P(X=0, Y=1) = 0.4$, $P(X=1, Y=0) = 0.2$, $P(X=1, Y=1) = 0.3$. Calculate the mutual information $I(X; Y)$.

   [6]

   c) Consider two probability distributions P and Q over the same set {a, b, c}: P(a)=0.5, P(b)=0.3, P(c)=0.2; Q(a)=0.4, Q(b)=0.4, Q(c)=0.2. Compute the relative entropy (i.e., Kullback-Leibler divergence) $D(P||Q)$.

   [6]

   d) Suppose a binary random variable X is transmitted over a noisy channel and Y is the received signal. The probability of decoding error is $P_e = 0.1$. Apply Fano's inequality to find an upper bound on the conditional entropy $H(X|Y)$.

   [7]

2. Rate-distortion theory (25 marks).

a) Consider the rate-distortion function $R(D) = \min I(x;\hat{x}), E_{x,\hat{x}} d(x,\hat{x}) \le D$, where $E_{x,\hat{x}}$ denotes the expectation with respect to $x, \hat{x}$. Justify each step in the following derivation of the rate-distortion function for a Bernoulli source

$$X = \{0,1\},\ p_X = \{1-p, p\}\ (p \le 1/2), \text{ and } d(x,\hat{x}) = x \oplus \hat{x}.$$

In the following, (1), (2), …, (6) are the step numbers.

If $D \ge p \Rightarrow R(D) \overset{(1)}{=} 0;$

If $D < p \le 1/2,$

$$I(x;\hat{x}) \overset{(2)}{=} H(x) - H(x \mid \hat{x}) \overset{(3)}{=} H(p) - H(x \oplus \hat{x} \mid \hat{x})$$
$$\overset{(4)}{\ge} H(p) - H(x \oplus \hat{x}) \overset{(5)}{\ge} H(p) - H(D) \ \Rightarrow R(D) \overset{(6)}{\ge} H(p) - H(D).$$

[9]

b) Consider a continuous random variable x with mean zero and variance σ^2 and squared-error distortion.

i) Show the lower bound

$$h(x) - \frac{1}{2}\log(2\pi e D) \le R(D).$$

[6]

ii) Show that the Gaussian distribution has the largest rate-distortion function, i.e., the rate-distortion function for x is upper bounded by

$$R(D) \le \frac{1}{2}\log\frac{\sigma^2}{D}.$$

Hint: use the following joint distribution of x and $\hat{x}$ in Fig. 2.1.

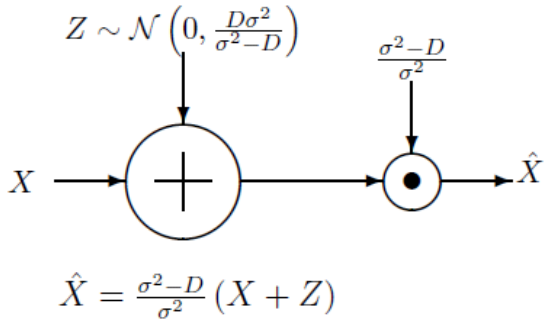


Fig. 2.1. Joint distribution of x and $\hat{x}$. x and z are independent.

[10]

3. Channel capacity (25 marks).

a) Calculate the capacity of the following channels with probability transition matrix

i) $Q = \begin{bmatrix} 1/2 & 1/6 & 1/3 \\ 1/6 & 1/2 & 1/3 \end{bmatrix} \quad x \in \{0,1\} \quad y \in \{0,1,2\}$

ii) $Q = \begin{bmatrix} 1/2 & 1/3 & 1/6 \\ 1/6 & 1/2 & 1/3 \\ 1/3 & 1/6 & 1/2 \end{bmatrix} \quad x, y \in \{0,1,2\}$

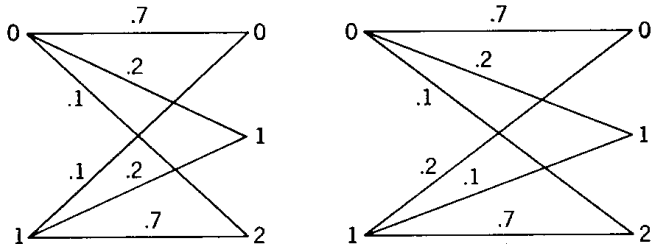
iii) $Q = \begin{bmatrix} 1/2 & 1/3 & 1/6 & 0 \\ 0 & 1/2 & 1/3 & 1/6 \\ 1/6 & 0 & 1/2 & 1/3 \\ 1/3 & 1/6 & 0 & 1/2 \end{bmatrix} \quad x, y \in \{0,1,2,3\}$

[9]

b) In fact, the property that the uniform input distribution achieves the capacity of weakly symmetric channels can be extended to *generally symmetric channels* (to be proved in part c). In these channels, the set of outputs can be partitioned into subsets in a way, for each subset of the transition matrix: all rows are permutations of each other, and all columns are permutations of each other (similar to symmetric channels).

i) Show that the binary erasure channel is a generally symmetric channel, but is not a weakly symmetric channel.

ii) Which one of the following two channels is generally symmetric? Compute the capacity of the generally symmetric channel.



(1) (2)

Fig. 3.1. Transition probabilities of two discrete memoryless channels (DMC).

[8]

c) Now, let us prove the uniform input distribution achieves the capacity of *generally symmetric channels*, in two steps.

i) Firstly, using the method of Lagrange multipliers, show that for any DMC, if there is an input distribution $Q(x) > 0$ for all x, such that

$$I(x;Y) = C \qquad \text{for all } x \tag{3.1}$$

where

$$I(x;Y) = \sum_y P(y|x) \log \frac{P(y|x)}{\sum_x Q(x)P(y|x)},$$

then C is the channel capacity.

ii) Show that Equation (3.1) holds for generally symmetric channels if the input distribution is uniform.

[8]

4. Network information theory (25 marks).

a) Slepian-Wolf coding. Let (X, Y) have the joint probability mass function

$p(x,y)$	0	1
0	1/8	1/4
1	3/8	1/4

Calculate and sketch the Slepian-Wolf rate region for this source pair.

[10]

b) Consider the following degraded broadcast channel, where Y_1 and Y_2 are two receivers, and E denotes Erasure.

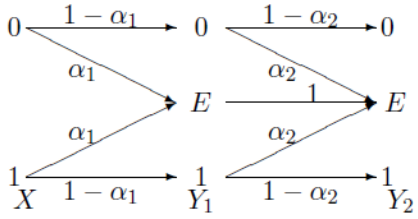


Fig. 4.1. Degraded broadcast channel, where E denotes Erasure.

i) What is the capacity of the channel from X to Y_1?

[5]

ii) What is the capacity of the channel from X to Y_2?

[10]